Solution Architecture

Date	15 March 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

1. High-Level Architectural Pattern

CrimeVision AI adopts a **Layered Modular Architecture** with decoupled presentation, API routing, inference services, and persistent storage layers. This guarantees modular scalability, high testability, and isolated failure domains.

2. Architectural Layers & Component Breakdown

Architectural Layer	Components & Technologies	Responsibilities & Communication
Presentation Layer (Client Browser)	• Single Page Application (SPA) • Three.js WebGL 3D Canvas • Vanilla ES6 JS Controllers • FontAwesome 6 Icons • Cyberpunk Dark CSS	Renders responsive forensic HUD, handles user file drag-and-drop, interactive 3D camera controls, and asynchronous API communication via fetch() in api.js.
API & Routing Layer (Flask Blueprints)	• processing_bp (/api/process) • audio_analysis_bp (/api/analyze-audio) • video_analysis_bp (/api/analyze-video) • chat_bp (/api/chat) • results_bp (/api/reports)	Validates incoming multipart/form-data and JSON payloads, enforces file size constraints, sanitizes filenames, and routes to backend domain services.
AI Services Layer (Core Engine)	• vision_service (RT-DETR, CLIP) • audio_service (Whisper ASR) • video_service (1Hz sampler) • fusion_service (Cross-modal) • chat_service (Groq LLM) • pdf_service (ReportLab)	Executes deep learning inference, extracts features, calculates 3D coordinate meshes, synthesizes multilingual text, and constructs PDF reports.
Persistence & Storage (Dual-Tier Store)	• SessionStore (In-memory dict) • Disk Cache (JSON files) • Reports Vault (/data/reports) • Uploads Vault (/data/uploads)	Maintains resilient state caching, historical session retrieval by session_id, and static serving of evidence files and generated court PDFs.